

FUKUOKA, Japan

WMO: 478070

Lat: 33.5822N

Lon: 130.3764E

Elev: 15

StdP: 101.14

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.0	2.0	-8.5	1.8	4.0	-7.2	2.0	4.6	8.8	4.9	8.1	4.7	3.1	140	0.391

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.2	33.9	25.9	33.0	25.6	32.0	25.4	27.0	31.8	26.4	31.1	25.9	30.5	4.3	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.7	21.0	29.8	25.1	20.2	29.4	24.6	19.6	29.0	85.2	31.9	82.5	31.2	80.3	30.5	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.2	7.2	6.3	-1.0	35.8	1.2	0.9	-1.8	36.4	-2.5	36.9	-3.2	37.4	-4.0	38.0		
			-3.0	27.6	1.2	0.8	-3.9	28.2	-4.6	28.7	-5.3	29.2	-6.2	29.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.6	6.9	7.9	10.9	15.6	20.2	23.6	27.9	28.9	25.0	20.0	14.3	9.1
	DBStd	7.96	2.53	2.96	3.02	2.88	2.33	2.22	2.43	1.98	2.47	2.56	3.02	2.84
	HDD10.0	252	102	73	24	1	0	0	0	0	0	0	2	51
	HDD18.3	1396	356	293	229	88	8	0	0	0	0	12	123	286
	CDD10.0	3022	4	13	53	170	317	407	554	587	452	309	132	24
	CDD18.3	1124	0	0	0	8	67	157	296	328	202	62	4	0
	CDH23.3	10047	0	0	1	23	257	864	3263	3927	1511	197	4	0
	CDH26.7	3667	0	0	0	2	29	189	1332	1712	384	18	0	0
Wind	WSAvg	2.8	2.8	2.8	3.0	2.9	2.7	2.6	2.8	2.8	2.9	2.6	2.5	2.7
Precipitation	PrecAvg	1647	73	70	100	125	142	254	266	184	180	95	87	64
	PrecMax	2980	208	155	224	240	496	642	886	847	609	290	182	148
	PrecMin	895	13	18	21	35	37	15	19	8	24	11	13	16
	PrecStd	369	38	33	42	52	80	133	182	144	110	63	44	31
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.2	19.2	21.9	25.7	29.1	32.1	34.6	35.7	33.3	28.5	23.9	18.4
		MCWB	12.6	14.7	14.2	17.9	19.5	24.4	26.0	26.1	25.1	22.4	18.0	13.7
	2%	DB	13.6	16.3	19.4	23.7	27.3	30.1	33.6	34.5	31.5	26.8	21.7	16.2
		MCWB	10.0	12.4	13.6	16.3	18.8	23.3	25.8	26.0	24.8	21.0	16.9	12.0
	5%	DB	12.2	14.1	17.7	22.1	26.0	28.8	32.7	33.5	30.1	25.4	20.4	14.9
		MCWB	8.6	10.3	12.8	15.6	18.3	22.7	25.6	25.8	24.1	20.0	15.9	11.0
	10%	DB	11.0	12.5	16.0	20.6	24.7	27.5	31.8	32.5	28.8	24.2	19.2	13.7
		MCWB	7.6	8.9	11.6	14.7	17.8	22.0	25.4	25.6	23.6	19.0	15.1	10.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.5	15.3	16.7	20.2	22.9	25.3	27.7	27.9	26.2	24.1	19.9	15.4
		MCDB	15.3	18.0	18.9	23.2	26.0	29.7	32.4	33.2	30.4	26.8	21.9	16.8
	2%	WB	11.2	13.3	14.8	18.2	21.4	24.5	26.9	27.2	25.5	22.5	18.1	13.2
		MCDB	12.6	15.9	17.8	21.5	24.4	28.5	31.6	32.1	29.6	25.7	20.4	14.9
	5%	WB	9.2	11.2	13.4	17.0	20.2	23.9	26.3	26.7	24.9	21.3	17.0	11.8
		MCDB	11.4	13.3	16.7	20.4	23.6	27.6	30.9	31.4	28.8	24.3	19.7	14.1
	10%	WB	7.9	9.5	12.2	15.9	19.2	23.1	25.8	26.2	24.3	19.9	15.7	10.5
		MCDB	10.6	12.0	15.5	19.3	23.1	26.4	30.3	30.8	28.0	22.9	18.7	13.4
Mean Daily Temperature Range	5% DB	MDBR	5.6	6.2	7.0	7.5	7.4	6.1	5.9	6.2	6.1	6.8	6.6	6.0
		MCDBR	7.7	8.6	9.3	9.6	9.4	7.6	6.9	7.4	7.1	7.9	7.8	7.6
		MCWBR	5.4	6.1	6.1	5.4	4.7	3.5	2.8	2.9	3.2	4.1	4.7	5.2
	5% WB	MCDBR	6.8	7.2	7.8	7.3	7.0	6.1	6.5	6.7	6.2	6.3	6.5	6.4
		MCWBR	5.9	6.2	6.0	4.9	4.8	3.2	2.9	3.0	3.1	3.8	4.7	5.5
Clear-Sky Solar Irradiance	taub	0.432	0.473	0.534	0.554	0.562	0.577	0.568	0.536	0.506	0.460	0.432	0.424	
	taud	2.051	1.977	1.832	1.838	1.880	1.927	1.996	2.085	2.118	2.145	2.137	2.086	
	Ebn at Noon	745	758	744	753	752	738	742	760	763	764	743	726	
	Edh at Noon	134	160	200	207	201	191	177	160	149	134	122	123	
All-Sky Solar Radiation	RadAvg	2.17	2.88	3.97	4.80	5.22	4.45	4.86	5.11	4.11	3.48	2.58	2.01	
	RadStd	0.20	0.28	0.27	0.41	0.60	0.55	0.72	0.56	0.42	0.36	0.25	0.14	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	+0.55	N/A	N/A	+80	+62	
(47)	Regional (28 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page